

# C. D. ALIPRANTIS'S QUESTIONS ON LOCALLY SOLID TOPOLOGIES

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ABSTRACT. ToDo

## 1. INTRODUCTION

C. D. Aliprantis's early work focused on locally solid vector lattices [14]. This line of research culminated in the publication of the standard monograph in the field, coauthored with O. Burkinshaw [3]. In one of his earliest papers, published in 1974, Aliprantis studied completions of locally solid vector lattices [2]. At the end of that paper, he posed a list of open questions whose resolution would help complete the theory. The goal of this note is to answer all these questions in the negative by constructing counterexamples and to formalize them in Lean. For each question, we shall also provide enough context to motivate it and show how it fits into the theory.

Next, we give an overview of the questions considered. The questions in Sections 2 to 4 and 6 concern properties or implications known to hold in the metrizable case, and Aliprantis asked whether they also hold in the non-metrizable case. More precisely, Sections 2 and 3 ask whether the  $\sigma$ -Lebesgue and (B, i) properties, respectively, pass to the completion in the non-metrizable case. On the other hand, Sections 4 and 6 ask whether the metrizability assumptions in Theorems 4.1 and 6.1, respectively, can be dropped. Finally, the question in Section 5 is slightly different: it asks whether the condition in Theorem 5.1, which characterizes when the image of a Hausdorff locally solid vector lattice in its completion is regular, can be weakened.

Aliprantis, together with O. Burkinshaw, posed two further questions on locally solid vector lattices in a later paper [1]. Although they are still listed as open questions in [14], their consistency was actually clarified by D. H. Fremlin [6], G. Buskes, and I. Labuda [5], who showed that an affirmative answer to either question is equiconsistent with the nonexistence of measurable cardinals.

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**1.1. Preliminaries and notation.** For terminology concerning vector lattices (also called Riesz spaces) and locally solid topologies that is not introduced here, we refer the reader to the monograph [3]. In the remainder of this section, we recall some standard facts and notation that will be used frequently.

Given a vector space  $X$  and a family of seminorms  $\mathcal{P}$  on  $X$ , the topology generated by  $\mathcal{P}$  is the unique locally convex linear topology for which the sets of the form

$$\{x \in X : p_1(x), \dots, p_n(x) < \varepsilon\},$$

where  $n \in \mathbb{N}$ ,  $p_1, \dots, p_n \in \mathcal{P}$ , and  $\varepsilon > 0$ , form a basis of neighborhoods of zero. This topology is Hausdorff if and only if, for every nonzero  $x \in X$ , there exists  $p \in \mathcal{P}$  such that  $p(x) \neq 0$ . If  $X$  is a vector lattice and every element of  $\mathcal{P}$  is a lattice seminorm, then  $X$ , equipped with the topology generated by  $\mathcal{P}$ , is a locally convex-solid vector lattice.

Throughout, a Cauchy net is a topologically Cauchy net; that is, a net  $(x_\alpha) \subseteq X$  such that, for every neighborhood of zero  $U$ , there exists an index  $\alpha_0$  for which  $x_\alpha - x_\beta \in U$  for all  $\alpha, \beta \geq \alpha_0$ . Similarly, a convergent net is always understood to be topologically convergent, and  $x_\alpha \rightarrow x$  always denotes topological convergence. A topological vector space is topologically complete if every Cauchy net converges.

Let  $X$  be a Hausdorff topological vector space. Suppose that there is a complete Hausdorff topological vector space  $\hat{X}$  and an embedding  $J: X \rightarrow \hat{X}$  (that is, a linear topological embedding) such that  $J(X)$  is dense in  $\hat{X}$ . Then the pair  $(\hat{X}, J)$  is unique up to isomorphism and is called the topological completion of  $X$ . Although  $X$  can be identified with  $J(X)$ , and hence regarded as a linear subspace of  $\hat{X}$ , we will usually display  $J$  explicitly to avoid ambiguity.

Every Hausdorff topological vector space admits a topological completion [13, Section 1.5]. If  $X$  is a Hausdorff locally solid vector lattice, then the closure of  $J(X_+)$  in  $\hat{X}$  equips  $\hat{X}$  with the structure of a Hausdorff locally solid vector lattice and makes  $J$  a lattice homomorphism. Henceforth, when we refer to the completion of a Hausdorff locally solid vector lattice, we mean this particular Hausdorff locally solid vector lattice structure.

In practice, to verify that a Hausdorff locally solid vector lattice  $Y$ , together with a map  $J: X \rightarrow Y$ , is the completion of  $X$ , it suffices to check the following:

- (i)  $Y$  is complete.
- (ii)  $J$  is a vector lattice homomorphism and a topological embedding.
- (iii)  $J(X_+)$  is dense in  $Y_+$ .

The third condition ensures that  $Y_+$  is the closure of  $J(X_+)$  and that  $J(X)$  is dense in  $Y$ . This checklist will recur frequently.

## 1.2. Lean formalization.

## 2. QUESTION 1

**2.1. Background.** We begin by recalling two basic definitions in the theory of locally solid vector lattices.

**Definition 2.1.** Let  $X$  be a locally solid vector lattice.

- (i) We say that  $X$  satisfies the  $\sigma$ -Lebesgue property if, for every sequence  $(x_n) \subseteq X_+$ ,  $x_n \downarrow 0$  implies  $x_n \rightarrow 0$ .
- (ii) We say that  $X$  satisfies the Lebesgue property if, for every net  $(x_\alpha) \subseteq X_+$ ,  $x_\alpha \downarrow 0$  implies  $x_\alpha \rightarrow 0$ .

These properties were called (A, i) and (A, ii), respectively, in [10] and in the reference paper [2]. We use instead the terminology of [3], which originates in a paper by D. H. Fremlin [7]; see the historical note following [3, Definition 3.1]. These two properties have been studied extensively [3, Chapter 3], especially when  $X$  is a Banach lattice and  $\tau$  is the topology induced by the norm—that is, when  $X$  is a  $(\sigma)$ -order continuous Banach lattice [15]. In [2], Aliprantis investigated whether the  $(\sigma)$ -Lebesgue property passes to the topological completion and obtained the following results.

**Theorem 2.2** ([2, Theorem 3.2] and [2, Theorem 5.1]). *Let  $X$  be a Hausdorff locally solid vector lattice.*

- (i) *If  $X$  is Lebesgue, then its topological completion is also Lebesgue.*
- (ii) *If  $X$  is metrizable and  $\sigma$ -Lebesgue, then its topological completion is also  $\sigma$ -Lebesgue.*

These results motivate the following natural question.

**Question 1** ([2, Open Problem 1]). *Is the  $\sigma$ -Lebesgue property preserved under topological completion without assuming metrizability?*

**2.2. Solution.** We construct a Hausdorff locally solid vector lattice that is  $\sigma$ -Lebesgue but whose topological completion is not  $\sigma$ -Lebesgue.

Let  $A$  be an uncountable set with the discrete topology, and let  $A_\infty = A \cup \{\infty\}$  be its one-point compactification. Recall that a function  $f: A_\infty \rightarrow \mathbb{R}$  is continuous if and only if, for every  $\varepsilon > 0$ , the set

$$\{a \in A : |f(a) - f(\infty)| \geq \varepsilon\}$$

is finite. We equip  $C(A_\infty)$  with the topology whose convergence is uniform on a countable subset of  $A$  and pointwise everywhere else. More precisely, fix a countable subset  $B = \{b_j : j \in \mathbb{N}\} \subseteq A$ , denote  $A' = A \setminus B$ , and define

$$q(f) = \sup_j |f(b_j)| \quad \text{and} \quad q_a(f) = |f(a)|$$

for every  $a \in A'$  and  $f \in C(A_\infty)$ . The functions  $q$  and  $q_a$ , for  $a \in A'$ , are lattice seminorms on  $C(A_\infty)$ . Let  $\tau$  be the locally convex-solid topology on  $C(A_\infty)$  generated by  $\{q\} \cup \{q_a : a \in A'\}$ . This topology is Hausdorff: if  $f \in C(A_\infty)$  is nonzero at some  $a \in A'$ , then  $q_a(f) \neq 0$ ; if it is nonzero at some  $b_j \in B$ , then  $q(f) \neq 0$ ; and if  $f(\infty) \neq 0$ , then also  $q(f) \neq 0$  because  $f(b_j) \rightarrow f(\infty)$ .

**Lemma 2.3.** *The Hausdorff locally solid vector lattice  $(C(A_\infty), \tau)$  has the  $\sigma$ -Lebesgue property.*

*Proof.* Let  $(f_n) \subseteq C(A_\infty)$  be a decreasing sequence satisfying  $f_n \downarrow 0$ . We show that  $f_n \rightarrow 0$  in the  $\tau$  topology; equivalently, that  $q(f_n) \rightarrow 0$  and  $q_a(f_n) \rightarrow 0$  for every  $a \in A'$ .

First we show that  $f_n(a) \rightarrow 0$  for every  $a \in A$ . Suppose not. Then there would exist  $a \in A$  and  $\varepsilon > 0$  such that  $f_n(a) \geq \varepsilon$  for infinitely many  $n \in \mathbb{N}$ ; since  $(f_n(a))$  is a decreasing sequence, actually  $f_n(a) \geq \varepsilon$  for all  $n \in \mathbb{N}$ . But then  $\varepsilon \chi_{\{a\}}$  is a nonzero lower bound for  $f_n$ , contradicting the assumption. This shows, in particular, that  $q_a(f_n) = |f_n(a)| \rightarrow 0$  for every  $a \in A'$ .

We also have  $f_n(\infty) \rightarrow 0$ . Again, suppose this were not the case. Then there would exist an  $\varepsilon > 0$  such that  $f_n(\infty) \geq \varepsilon$  for all  $n \in \mathbb{N}$ . By continuity at  $\infty$ , for each  $n \in \mathbb{N}$  there exists a finite  $F_n \subseteq A$  such that

$$f_n(a) \geq \varepsilon/2 \quad \text{for all } a \in A \setminus F_n.$$

Since  $A$  is uncountable, we may choose  $a \in A \setminus \bigcup_{n \in \mathbb{N}} F_n$ . Then  $(\varepsilon/2)\chi_{\{a\}}$  is a nonzero lower bound of  $(f_n)$ , again contradicting the assumption.

Thus we have a decreasing sequence  $(f_n)$  of continuous functions that converges pointwise to 0 on the compact space  $A_\infty$ . By Dini's theorem,  $(f_n)$  converges uniformly to 0. In particular,  $q(f_n) = \sup_j |f_n(b_j)|$  tends to 0.  $\square$

Let  $c$  denote the space of real convergent sequences, with its usual Banach lattice structure. We next show that  $c \times \mathbb{R}^{A'}$ , with its natural structure as the product of two Hausdorff locally solid vector lattices, is the completion of  $(C(A_\infty), \tau)$ .

**Lemma 2.4.** *The Hausdorff locally solid vector lattice  $c \times \mathbb{R}^{A'}$ , together with the embedding*

$$\begin{aligned} J: \quad C(A_\infty) &\longrightarrow c \times \mathbb{R}^{A'} \\ f &\longmapsto ((f(b_j))_{j \in \mathbb{N}}, (f(a))_{a \in A'}) \end{aligned}$$

*is the topological completion of  $(C(A_\infty), \tau)$ .*

*Proof.* Recall that convergence in  $c \times \mathbb{R}^{A'}$  is uniform on  $c$  and pointwise on  $\mathbb{R}^{A'}$ . This space is topologically complete. The map  $J$  is a vector lattice homomorphism, and it is injective because  $f(b_j) \rightarrow f(\infty)$ . Moreover, by the definition of  $\tau$ ,  $J$  is a homeomorphism onto its image.

It only remains to show that  $J(C(A_\infty)_+)$  is dense in  $(c \times \mathbb{R}^{A'})_+$ . Let  $g = (g_1, g_2) \in (c \times \mathbb{R}^{A'})_+$ , and let  $U$  be a neighborhood of  $g$ . There exist  $\varepsilon > 0$ ,  $m \in \mathbb{N}$ , and  $a_1, \dots, a_m \in A'$  such that

$$\{ h = (h_1, h_2) \in c \times \mathbb{R}^{A'} : \|g_1 - h_1\|_\infty < \varepsilon, |g_2(a_i) - h_2(a_i)| < \varepsilon \text{ for } i = 1, \dots, m \}$$

is contained in  $U$ . Define  $f: A_\infty \rightarrow \mathbb{R}$  by

$$\begin{aligned} f(b_j) &= g_1(j) \quad \text{for } j \in \mathbb{N}, \\ f(a_i) &= g_2(a_i) \quad \text{for } i = 1, \dots, m, \\ f(a) &= \lim_j g_1(j) \quad \text{for } a \in A' \setminus \{a_1, \dots, a_m\}, \\ f(\infty) &= \lim_j g_1(j). \end{aligned}$$

Then  $f \in C(A_\infty)_+$ . Moreover, the first coordinate of  $J(f)$  agrees with  $g_1$ , while the second agrees with  $g_2$  on  $\{a_1, \dots, a_m\}$ . Hence  $J(f) \in U$ .  $\square$

However,  $c \times \mathbb{R}^{A'}$  does not have the  $\sigma$ -Lebesgue property. Indeed, for  $n \in \mathbb{N}$ , let  $x_n \in c$  be the sequence whose first  $n$  entries are 0 and whose remaining entries are 1. Then  $x_n \downarrow 0$ , but  $\|x_n\|_\infty = 1$  for every  $n \in \mathbb{N}$ . Consequently,  $(x_n, 0) \downarrow 0$  in  $c \times \mathbb{R}^{A'}$ , but this sequence does not converge to 0 in the product topology. In summary, we have proved the following.

**Theorem 2.5.** *There exists a Hausdorff  $\sigma$ -Lebesgue locally convex-solid vector lattice whose topological completion is not  $\sigma$ -Lebesgue.*

### 3. QUESTION 2

**3.1. Background.** Aliprantis introduced the following property in [2, Section 1] as a generalization of the analogous property for normed vector spaces introduced by W. A. J. Luxemburg and A. C. Zaanen in [11, Section 34].

**Definition 3.1.** We say that a locally solid vector lattice  $X$  satisfies *property (B, i)* if every sequence  $(x_n) \subseteq X_+$  that is topologically bounded and increasing is Cauchy.

When  $X$  is a Banach lattice equipped with its norm topology, property (B, i) is equivalent to  $X$  being a KB-space: every increasing, norm-bounded sequence in  $X_+$  converges. Property (B, i) was introduced together with its analogue for nets, property (B, ii). They are the topologically bounded counterparts of properties (A, iii) and (A, iv), respectively, which require every order-bounded increasing sequence and net to be Cauchy. The latter properties were first introduced for normed vector lattices [10, Section 33] to prove results related to a theorem of H. Nakano. Aliprantis then showed that property (B, i) is preserved under topological completion for metrizable locally solid vector lattices.

**Theorem 3.2** ([2, Theorem 5.3]). *A metrizable locally solid vector lattice has property (B, i) if and only if its topological completion has property (B, i).*

Clearly, if the completion satisfies property (B, i), then so does the original space. This leaves the converse implication open in the non-metrizable case.

**Question 2** ([2, Open Problem 1]). *Is property (B, i) preserved under topological completion without assuming metrizability?*

**3.2. Solution.** We construct a Hausdorff locally solid vector lattice with property (B, i) whose completion does not have property (B, i). It will be a vector sublattice of  $c_0 \times \mathbb{R}^{PS}$ , where  $PS = (\mathbb{R}_+)^{\mathbb{N}}$  is the space of nonnegative sequences and  $c_0$  is the Banach lattice of real sequences converging to zero, equipped with the uniform norm. With the product topology,  $c_0 \times \mathbb{R}^{PS}$  is a complete Hausdorff locally convex-solid vector lattice.

For  $j \in \mathbb{N}$ , define  $a_j = ((a_j)_1, (a_j)_2) \in c_0 \times \mathbb{R}^{PS}$  by  $(a_j)_1(m) = \delta_{jm}$ , where  $\delta_{jm}$  denotes the Kronecker delta, and  $(a_j)_2(\alpha) = \alpha(j)$ . Let  $Y$  be the vector sublattice generated by  $\{a_j : j \in \mathbb{N}\}$ . Let

$$I = \{(0, f) \in c_0 \times \mathbb{R}^{PS} : f \text{ is finitely supported}\}$$

be the ideal of finitely supported functions on  $PS$ , extended by zero on  $\mathbb{N}$ . This is an ideal of  $c_0 \times \mathbb{R}^{PS}$ , so  $X = Y + I$  is a vector sublattice of  $c_0 \times \mathbb{R}^{PS}$ . Equipped with the subspace topology,  $X$  is a Hausdorff locally convex-solid vector lattice.

**Lemma 3.3.** *The topological completion of  $X$  is  $c_0 \times \mathbb{R}^{PS}$ , together with the natural inclusion.*

*Proof.* It only remains to show that  $X_+$  is dense in  $(c_0 \times \mathbb{R}^{PS})_+$ . Let  $z = (z_1, z_2) \in (c_0 \times \mathbb{R}^{PS})_+$ , and let  $U$  be an open neighborhood of  $z$ . Then there exist  $\varepsilon > 0$ ,  $m \in \mathbb{N}$ , and  $\alpha_1, \dots, \alpha_m \in PS$  such that

$$\{w = (w_1, w_2) \in c_0 \times \mathbb{R}^{PS} : \|z_1 - w_1\|_\infty < \varepsilon, |z_2(\alpha_i) - w_2(\alpha_i)| < \varepsilon \text{ for } i = 1, \dots, m\}$$

is contained in  $U$ . By retaining only finitely many coordinates of  $z_1$ , choose a positive, finitely supported sequence  $y'$  such that  $\|y' - z_1\|_\infty < \varepsilon$ . Let  $y = \sum_{j \in \mathbb{N}} y'(j)a_j \in Y$ , and define  $(0, f) \in I$  by

$$f(\alpha_i) = z_2(\alpha_i) - \sum_{j \in \mathbb{N}} y'(j)\alpha_i(j)$$

for  $i = 1, \dots, m$ , and  $f(\alpha) = 0$  otherwise. Then  $x = y + (0, f) \in X_+$ . If  $x = (x_1, x_2)$ , then  $\|x_1 - z_1\|_\infty = \|y' - z_1\|_\infty < \varepsilon$  and  $x_2(\alpha_i) = z_2(\alpha_i)$  for  $i = 1, \dots, m$ . Thus  $x \in U$ .  $\square$

Since  $c_0$  is not a KB-space, the space  $c_0 \times \mathbb{R}^{PS}$  does not have property (B, i).

**Lemma 3.4.**  *$c_0 \times \mathbb{R}^{PS}$  does not have property (B, i).*

*Proof.* For  $n \in \mathbb{N}$ , let  $s_n \in c_0$  be the sequence whose first  $n$  entries are 1 and whose remaining entries are 0. Then the sequence  $(s_n, 0)$  in  $c_0 \times \mathbb{R}^{PS}$  is positive, increasing, and topologically bounded, but it is not Cauchy.  $\square$

The difficult part is to show that  $X$  has property (B, i).

**Lemma 3.5.**  *$X$  has property (B, i).*

*Proof.* Let  $(x_n) \subseteq X_+$  be an increasing, topologically bounded sequence. We need to show that it is Cauchy. Write

$$x_n = y_n + (0, z_n) = ((x_n)_1, (x_n)_2),$$

where  $y_n = ((y_n)_1, (y_n)_2) \in Y$  and  $(0, z_n) \in I$ . Thus  $(x_n)_1 = (y_n)_1$  and  $(x_n)_2 = (y_n)_2 + z_n$ . Notice that  $((x_n)_2)$  is immediately Cauchy in the product topology of  $\mathbb{R}^{PS}$ : for every  $\alpha \in PS$ , the sequence  $((x_n)_2(\alpha))$  is increasing and bounded, hence Cauchy. Thus it only remains to show that  $((x_n)_1)$  is Cauchy in the uniform norm.

Since  $((x_n)_1)$  is uniformly bounded, the sequence

$$u(k) = \sup_n (x_n)_1(k) \quad (k \in \mathbb{N})$$

belongs to  $\ell_\infty$ . If  $((x_n)_1)$  is Cauchy, then  $u$  is the limit of this sequence in the uniform norm, and therefore belongs to  $c_0$ . Conversely, if  $u \in c_0$ , then it follows from Dini's theorem that  $((x_n)_1)$  converges uniformly to  $u$  (when seen as elements defined on the one-point compactification of  $\mathbb{N}$ ), and therefore that it is Cauchy. In summary,  $((x_n)_1)$ , and in fact  $(x_n)$ , is Cauchy if and only if  $u \in c_0$ .

Suppose, to reach a contradiction, that  $(x_n)$  is not Cauchy. Then  $u \notin c_0$ . Let

$$E = \bigcup_{n \in \mathbb{N}} \text{supp } z_n \subseteq PS.$$

This set is countable, and for every  $\alpha \in PS \setminus E$ , we have  $(x_n)_2(\alpha) = (y_n)_2(\alpha)$  for all  $n \in \mathbb{N}$ . We are going to use the fact that  $u \notin c_0$  to construct a point  $\alpha \in PS \setminus E$  and a subsequence  $(x_{n_j})$  such that

$$(x_{n_j})_2(\alpha) = (y_{n_j})_2(\alpha) > j \quad (j \in \mathbb{N}),$$

thus contradicting the topological boundedness of  $(x_n)$ . Since  $u \notin c_0$ , there is  $\varepsilon > 0$  such that, for every finite  $F \subseteq \mathbb{N}$ , some  $k \in \mathbb{N} \setminus F$  satisfies  $u(k) > \varepsilon$ . Consequently, for every finite  $F \subseteq \mathbb{N}$  and every  $N \in \mathbb{N}$ , there are  $k \in \mathbb{N} \setminus F$  and  $n \geq N$  such that  $(x_n)_1(k) > \varepsilon$ .

For every  $n \in \mathbb{N}$ , choose a finite set  $S_n \subseteq \mathbb{N}$  and a lattice-linear expression for  $y_n$  in terms of  $\{a_i : i \in S_n\}$ . In particular,  $(y_n)_1$  is supported on  $S_n$ . We are going to use the preceding observation inductively to construct strictly increasing sequences  $(n_j)$  and  $(m_j)$  such that

$$(x_{n_j})_1(m_j) = (y_{n_j})_1(m_j) > \varepsilon \quad \text{and} \quad m_j \notin \bigcup_{i < j} S_{n_i}.$$

The construction is straightforward. Having chosen  $n_i$  and  $m_i$  for  $i < j$ , let

$$H_{j-1} = \bigcup_{i < j} S_{n_i} \cup \{1, \dots, m_{j-1}\},$$

where  $m_0 = 1$ . Applying the preceding observation with  $F = H_{j-1}$  and  $N = n_{j-1} + 1$ , choose  $m_j \notin H_{j-1}$  and  $n_j > n_{j-1}$  such that  $(x_{n_j})_1(m_j) > \varepsilon$ . Since  $(y_{n_j})_1$  is supported on  $S_{n_j}$ , we have  $m_j \in S_{n_j}$ .

If  $E$  is nonempty, write  $E = \{\beta^{(j)} : j \in \mathbb{N}\}$ , allowing repetitions if  $E$  is finite. We now construct  $\alpha \in PS$ . This sequence will be supported on  $\{m_j : j \in \mathbb{N}\}$ , and its values will be chosen in order: at stage  $j$ , all coordinates in  $\bigcup_{i < j} S_{n_i}$  have already been assigned. Let  $\Phi_j : \mathbb{R}^{S_{n_j}} \rightarrow \mathbb{R}$  be the lattice-linear expression defining  $y_{n_j}$ , and let  $w_j \in \mathbb{R}^{S_{n_j}}$  agree with the values of  $\alpha$  already assigned on  $S_{n_j} \cap \bigcup_{i < j} S_{n_i}$  and be zero elsewhere. If  $e_{m_j}$  denotes the standard unit vector at  $m_j$ , then

$$\Phi_j(e_{m_j}) = (y_{n_j})_1(m_j) > \varepsilon.$$

By positive homogeneity and continuity,

$$\frac{\Phi_j(w_j + te_{m_j})}{t} = \Phi_j\left(\frac{w_j}{t} + e_{m_j}\right) \longrightarrow \Phi_j(e_{m_j}) \quad \text{as } t \rightarrow \infty.$$

Hence we may choose  $t_j > 0$  such that  $\Phi_j(w_j + t_j e_{m_j}) > j$  and, if  $E$  is nonempty,  $t_j \neq \beta^{(j)}(m_j)$ . Set  $\alpha(m_j) = t_j$  and assign the value 0 to every still unassigned coordinate in  $S_{n_j} \setminus \{m_j\}$ . After all stages, assign the value 0 to every remaining coordinate.

The  $\alpha$  thus constructed certainly belongs to  $PS$ , and is such that  $\alpha \notin E$ , since  $\alpha$  differs from  $\beta^{(j)}$  at coordinate  $m_j$ , for each  $j \in \mathbb{N}$ . It should be clear from the construction that  $(y_{n_j})_2(\alpha) > j$ , because  $(y_{n_j})_2$  only depends on the values of  $\alpha$  at  $S_{n_j}$ , and is therefore equal to  $\Phi_j(w_j + t_j e_{m_j}) > j$ . Since  $\alpha \notin E$ ,  $(x_{n_j})_2(\alpha) = (y_{n_j})_2(\alpha) > j$ , thus showing that  $((x_n)_2)$  is not bounded on  $\mathbb{R}^{PS}$ .  $\square$

Thus we have established the following.

**Theorem 3.6.** *There exists a Hausdorff locally convex-solid vector lattice with property  $(B, i)$  whose topological completion does not have property  $(B, i)$ .*

#### 4. QUESTION 3

**4.1. Background.** Let  $X$  be a Hausdorff locally solid vector lattice, and let  $\hat{X}$  be its topological completion. We call an element  $\hat{x} \in \hat{X}$  an *upper element* if there exists an increasing sequence  $(x_n) \subseteq X_+$  converging to  $\hat{x}$ . The following result was first proved for normed vector lattices by W. A. J. Luxemburg [9, Theorem 60.3] and later extended by Aliprantis to metrizable locally solid vector lattices.

**Theorem 4.1** ([2, Lemma 4.1]). *Let  $X$  be a metrizable locally solid vector lattice, and let  $\hat{X}$  be its topological completion. Then every element of  $\hat{X}_+$  is the limit of a decreasing sequence of upper elements.*

**Question 3** ([2, Open Problem 2]). Does Theorem 4.1 remain true without assuming metrizability?



**4.2. Solution.** We construct a Hausdorff locally solid vector lattice whose completion contains a positive element that is not the limit of a decreasing sequence of upper elements.

The remaining counterexamples share a common flavor. We introduce some notation and facts that will be useful throughout. Let  $S$  be a topological space. We denote by  $C_k(S)$  the vector lattice  $C(S)$  of real-valued continuous functions on  $S$ , equipped with its usual pointwise vector lattice structure and the Hausdorff locally convex-solid topology induced by the family of seminorms

$$p_K(f) = \sup_{x \in K} |f(x)| \quad (f \in C_k(S)),$$

where  $K$  ranges over the compact subsets of  $S$ . This is the compact-open topology.

Whenever  $S$  and  $T$  are topological spaces, we write  $S \sqcup T$  for their disjoint union with its standard topology. We identify  $S$  and  $T$  with their canonical homeomorphic copies in  $S \sqcup T$ , so that an element of  $S \sqcup T$  is either an element of  $S$  or an element of  $T$ . With this convention, and the analogous convention for subsets and disjoint unions, every open subset of  $S \sqcup T$  has the form  $U \sqcup V$ , where  $U \subseteq S$  and  $V \subseteq T$  are open. Similarly, every compact subset of  $S \sqcup T$  has the form  $K \sqcup L$ , where  $K \subseteq S$  and  $L \subseteq T$  are compact. A map  $f: S \sqcup T \rightarrow R$ , where  $R$  is a topological space, is continuous if and only if both  $f|_S$  and  $f|_T$  are continuous. In that case, we write  $f = f|_S \sqcup f|_T$ .

**Proposition 4.2.** *Let  $M$  be a compact metric space, let  $P \subseteq M$ , and set  $Q = M \setminus P$ . Consider the injective map*

$$\begin{aligned} J: \quad C(M) &\longrightarrow C_k(P \sqcup Q) \\ f &\longmapsto f|_P \sqcup f|_Q \end{aligned}.$$

*Then  $C(M)$ , equipped with its usual pointwise vector lattice structure and the subspace topology induced by  $J$ , is a Hausdorff locally convex-solid vector lattice. Moreover,  $C_k(P \sqcup Q)$ , together with the embedding  $J$ , is the topological completion of  $C(M)$ .*

*Proof.* Since  $M$  is metrizable, so is  $P \sqcup Q$ ; hence  $C_k(P \sqcup Q)$  is complete [12, Theorem 5.8.7]. The map  $J$  is an injective vector lattice homomorphism. Identifying  $C(M)$  with  $J(C(M))$  and equipping it with the subspace topology makes it a Hausdorff locally convex-solid vector lattice, and  $J$  a topological embedding.

It remains to show that  $J(C(M)_+)$  is dense in  $C_k(P \sqcup Q)_+$ . Let  $f = f_P \sqcup f_Q \in C_k(P \sqcup Q)_+$ , and let  $K = K_P \sqcup K_Q$  be a compact subset of  $P \sqcup Q$ , where  $K_P \subseteq P$  and  $K_Q \subseteq Q$  are compact. The sets  $K_P$  and  $K_Q$  are disjoint compact subsets of the Hausdorff space  $M$ , hence they are closed in  $M$ . Therefore, the function  $h': K_P \cup K_Q \rightarrow \mathbb{R}$  defined by  $h'|_{K_P} = f_P|_{K_P}$  and  $h'|_{K_Q} = f_Q|_{K_Q}$  is positive and continuous. Since  $M$  is normal, the Tietze extension theorem provides a continuous function  $h \in C(M)$  that

agrees with  $h'$  on  $K_P \cup K_Q$ . Then  $J(|h|)|_K = f|_K$ , proving the density of  $J(C(M)_+)$ .  $\square$

With the notation of the preceding proposition, let  $M = \mathbb{R}$ ,  $P = \mathbb{R} \setminus \mathbb{Q}$ , and let  $J: C(M) \rightarrow C_k(P \sqcup Q)$  be the map defined above. We show that  $\chi_P \in C_k(P \sqcup Q)$  is not the limit of a decreasing sequence of upper elements.

**Lemma 4.3.** *Let  $f \in C_k(P \sqcup Q)$  be an upper element such that  $\chi_P \leq f$ . Then  $f(q) \geq 1$  for every  $q \in Q$ .*

*Proof.* Let  $(f_n) \subseteq C(\mathbb{R})_+$  be an increasing sequence converging to  $f$  in  $C_k(P \sqcup Q)$ . For  $t \in \mathbb{R}$ , set  $g(t) = \sup_n f_n(t)$ . Then  $g$  is lower semicontinuous on  $\mathbb{R}$ . Moreover, since convergence in  $C_k(P \sqcup Q)$  is pointwise, the restrictions of  $g$  to  $P$  and  $Q$  coincide with those of  $f$ .

Since  $f \geq \chi_P$ , we have  $g(t) \geq 1$  for all  $t \in P$ . Fix  $c < 1$ . By lower semicontinuity, the set  $g^{-1}(c, \infty)$  is open and contains  $P$ . We claim that  $g^{-1}(c, \infty) \cap \mathbb{Q}$  is dense in  $\mathbb{Q}$ . Indeed, every open interval contains an element of  $P$ , and hence meets  $g^{-1}(c, \infty)$ . This intersection is an open, nonempty subset of  $\mathbb{R}$ , so it contains a rational number. Since  $g|_Q = f|_Q$  is continuous and  $g^{-1}(c, \infty) \cap \mathbb{Q}$  is dense in  $\mathbb{Q}$ , we obtain  $f(q) = g(q) \geq c$  for every  $q \in \mathbb{Q}$ . As  $c < 1$  was arbitrary,  $f(q) \geq 1$  for every  $q \in \mathbb{Q}$ .  $\square$

Fix  $q_0 \in Q$ , and consider the open neighborhood of 0 given by

$$V = \{f \in C_k(P \sqcup Q) : |f(q_0)| < 1/2\}.$$

By the preceding lemma, every upper element  $f \geq \chi_P$  satisfies

$$(f - \chi_P)(q_0) = f(q_0) \geq 1.$$

Thus  $f \notin \chi_P + V$ . Consequently,  $\chi_P$  is not the limit of a decreasing sequence of upper elements: every element of such a sequence would be an upper element bounded below by  $\chi_P$ , whereas no such element belongs to the neighborhood  $\chi_P + V$  of  $\chi_P$ . We have therefore proved the following.

**Theorem 4.4.** *There exists a Hausdorff locally convex-solid vector lattice whose topological completion contains a positive element that is not the limit of a decreasing sequence of upper elements.*

## 5. QUESTION 4

**5.1. Background.** To understand how a Hausdorff locally solid vector lattice sits in its completion, it is useful to characterize when its image is regular. The following result was first proved in the locally convex setting by I. Kawai [8, Theorem 4.1] and later extended to the general setting by C. D. Aliprantis [2, Theorem 2.3].

**Theorem 5.1** ([3, Theorem 2.41]). *Let  $X$  be a Hausdorff locally solid vector lattice, and let  $\hat{X}$  be its topological completion with embedding  $J: X \rightarrow \hat{X}$ . Then the following are equivalent:*

- (i)  $J(X)$  is a regular vector sublattice of  $\hat{X}$ .

(ii) *Every topologically Cauchy net  $(x_\alpha) \subseteq X_+$  that order converges to zero in  $X$  also converges to zero in the topology.*

A natural weakening of condition (ii) is the requirement that every topologically Cauchy net  $(x_\alpha) \subseteq X_+$  satisfying  $x_\alpha \downarrow 0$  also satisfies  $x_\alpha \rightarrow 0$ . C. D. Aliprantis called this the *generalized  $(A, 0)$  property*; its sequential analogue is the  *$(A, 0)$  property*. He asked whether the generalized  $(A, 0)$  property is equivalent to condition (ii).

**Question 4** ([2, Open Problem 3]). Let  $X$  be a Hausdorff locally solid vector lattice satisfying the generalized  $(A, 0)$  property. Is its image in the topological completion a regular sublattice?

**5.2. Solution.** We construct a Hausdorff locally convex-solid vector lattice satisfying the generalized  $(A, 0)$  property whose image in its topological completion is not a regular sublattice.

Let  $C \subseteq [0, 1]$  be the middle-thirds Cantor set. Let  $P \subseteq C$  be any subset such that both  $P$  and  $C \setminus P$  are dense in  $C$  (for instance, take  $P$  to be the set of endpoints of the intervals removed in the construction of the Cantor set). Set  $Q = [0, 1] \setminus P$ , and let  $J: C[0, 1] \rightarrow C_k(P \sqcup Q)$  be the map from Proposition 4.2. Equip  $C[0, 1]$  with the subspace topology induced by  $J$ . Then it is a Hausdorff locally convex-solid vector lattice whose topological completion is  $C_k(P \sqcup Q)$ , together with the embedding  $J$ .

**Lemma 5.2.** *The vector sublattice  $J(C[0, 1])$  is not regular in  $C_k(P \sqcup Q)$ .*

*Proof.* Consider the set

$$D = \{ f \in C[0, 1]_+ : f \geq \chi_C \}.$$

Endow  $D$  with the reverse pointwise order. Then the net  $(f)_{f \in D}$  is decreasing in  $C[0, 1]$ . We show that  $\inf D = 0$ . Clearly, 0 is a lower bound for  $D$ . Let  $g \in C[0, 1]$  be any lower bound for  $D$ , and let  $t \in [0, 1] \setminus C$ . By Urysohn's lemma, there exists  $h \in C[0, 1]_+$  satisfying  $h \geq \chi_C$  and  $h(t) = 0$ . Since  $g \leq h$ , it follows that  $g(t) = 0$ . Thus  $g = 0$  on the dense subset  $[0, 1] \setminus C$ , and continuity yields  $g = 0$  on  $[0, 1]$ . Hence  $\inf D = 0$ . On the other hand,  $\chi_P$  is a nonzero lower bound for  $J(D)$ . Therefore,  $J(C[0, 1])$  is not regular in  $C_k(P \sqcup Q)$ .  $\square$

**Lemma 5.3.**  *$C[0, 1]$ , with the topology introduced above, has the generalized  $(A, 0)$  property.*

*Proof.* Let  $(f_\alpha) \subseteq C[0, 1]_+$  be a topologically Cauchy net such that  $f_\alpha \downarrow 0$ . Since  $C_k(P \sqcup Q)$  is complete and  $(J(f_\alpha))$  is also topologically Cauchy, there exists  $g \in C_k(P \sqcup Q)$  such that  $J(f_\alpha) \rightarrow g$ . It suffices to show that  $g = 0$ .

First we show that  $g|_Q = 0$ . Suppose, on the contrary, that  $g(q_0) > 0$  for some  $q_0 \in Q$ . Then there exist  $\varepsilon > 0$  and an open neighborhood  $U$  of  $q_0$  in  $[0, 1]$  such that  $g(q) > \varepsilon$  for every  $q \in U \cap Q$ . Since  $J(f_\alpha) \rightarrow g$  and  $(f_\alpha)$  is decreasing, we have  $f_\alpha(q) \geq g(q) > \varepsilon$  for every  $q \in U \cap Q$  and every  $\alpha$ . As  $[0, 1] \setminus C$  is dense and open, there is a nonempty open set  $V \subseteq U \setminus C$ .

By Urysohn's lemma, there exists a nonzero function  $h \in C[0, 1]_+$  such that  $0 \leq h \leq \varepsilon$  and  $h$  vanishes outside  $V$ . Then  $h \leq f_\alpha$  for every  $\alpha$ , contradicting  $f_\alpha \downarrow 0$ .

To finish the proof, we show that  $g|_P = 0$ . Suppose, on the contrary, that  $g(p_0) > 0$  for some  $p_0 \in P$ . Then there exist  $\varepsilon > 0$  and an open neighborhood  $U \subseteq [0, 1]$  of  $p_0$  such that  $g(p) > \varepsilon$  for all  $p \in U \cap P$ . Hence  $f_\alpha(p) > \varepsilon$  for every  $p \in U \cap P$  and every  $\alpha$ . Fix  $\alpha$ . Since  $f_\alpha$  is continuous on  $[0, 1]$  and  $P$  is dense in  $C$ , we have  $f_\alpha(t) \geq \varepsilon$  for all  $t \in U \cap C$ . As  $C \setminus P$  is dense in  $C$ , there exists  $s \in U \cap (C \setminus P)$ . Then  $s \in Q$  and  $f_\alpha(s) \geq \varepsilon$  for every  $\alpha$ , which implies  $g(s) = \inf_\alpha f_\alpha(s) \geq \varepsilon$ . This contradicts the fact that  $g|_Q = 0$ .  $\square$

In summary, we have proved the following.

**Theorem 5.4.** *There exists a Hausdorff locally convex-solid vector lattice satisfying the generalized  $(A, 0)$  property such that the image of this vector lattice in its topological completion is not regular.*

## 6. QUESTION 5

**6.1. Background.** Recall that a locally solid vector lattice  $X$  has property  $(A, 0)$  if every Cauchy sequence  $(x_n) \subseteq X_+$  satisfying  $x_n \downarrow 0$  also satisfies  $x_n \rightarrow 0$ . C. D. Aliprantis used the  $(A, 0)$  property to characterize regularity and order density for metrizable locally solid vector lattices in their topological completions.

**Theorem 6.1** ([2, Theorem 6.1]). *Let  $X$  be a metrizable locally solid vector lattice, let  $\hat{X}$  be its topological completion, and let  $J: X \rightarrow \hat{X}$  be the canonical embedding. Then the following are equivalent:*

- (i)  $X$  satisfies property  $(A, 0)$ .
- (ii)  $J(X)$  is order dense in  $\hat{X}$ .
- (iii)  $J(X)$  is regular in  $\hat{X}$ , i.e.,  $J$  preserves arbitrary suprema.
- (iv)  $J$  preserves countable suprema.
- (v) For every  $\hat{x} \in \hat{X}_+$ ,

$$\hat{x} = \sup\{Jx : x \in X, 0 \leq Jx \leq \hat{x}\}.$$

It is remarkable that order density of  $J(X)$  is equivalent to regularity. Aliprantis asked whether this remains true in the non-metrizable case.

**Question 5** ([2, Open Problem 4]). *Let  $X$  be a Hausdorff locally solid vector lattice, let  $\hat{X}$  be its topological completion, and let  $J: X \rightarrow \hat{X}$  be the canonical embedding. Is  $J(X)$  regular in  $\hat{X}$  if and only if it is order dense in  $\hat{X}$ ?*

**6.2. Solution.** We construct a Hausdorff locally solid vector lattice whose image in its completion is regular but not order dense. Since every order dense sublattice is regular, we are going to show that the nontrivial implication fails.

We reuse the counterexample from Section 4, including its notation. In fact, the following was already proved in [4, Example 9.8], thereby solving the open question without recognizing it as such. For completeness, we provide a detailed proof.

**Proposition 6.2** ([4, Example 9.8]).  *$J(C(\mathbb{R}))$  is regular in  $C_k(P \sqcup Q)$ , but not order dense.*

*Proof.* We first prove that  $J(C(\mathbb{R}))$  is regular. Let  $F \subseteq C(\mathbb{R})_+$  be such that  $J(F)$  has infimum zero in  $J(C(\mathbb{R}))$ . We show that  $J(F)$  has infimum zero in  $C_k(P \sqcup Q)$ . Suppose, on the contrary, that there exists  $h \in C_k(P \sqcup Q)_+$  such that

$$0 < h \leq J(f) \quad (f \in F).$$

Then  $h$  is strictly positive on a nonempty open subset of either  $P$  or  $Q$ . Since both  $P$  and  $Q$  are dense in  $\mathbb{R}$ , there are a nonempty open interval  $V \subseteq \mathbb{R}$  and  $\varepsilon > 0$  such that every  $f \in F$  satisfies  $f \geq \varepsilon$  on  $V$ . Choose a nonzero function  $\varphi \in C(\mathbb{R})_+$  such that  $\text{supp } \varphi \subseteq V$  and  $0 < \varphi \leq \varepsilon$ . Then

$$0 < J(\varphi) \leq J(f) \quad (f \in F),$$

contradicting the fact that  $J(F)$  has infimum zero in  $J(C(\mathbb{R}))$ . Hence  $J(C(\mathbb{R}))$  is regular in  $C_k(P \sqcup Q)$ .

But it is not order dense: there is no  $f \in C(\mathbb{R})_+$  such that  $0 < J(f) \leq \chi_P$ , since any such  $f$  would vanish on the rational numbers and therefore, by continuity, be identically zero.  $\square$

We have thus proved the following.

**Theorem 6.3.** *There exists a Hausdorff locally convex-solid vector lattice whose image in its topological completion is regular but not order dense.*

#### AI DISCLOSURE

The counterexamples in this paper were generated by GPT-5.5, and semi-autoformalized using GPT-5.6 Sol. This paper was entirely written by humans. GPT-5.6 Luna and Sol were used, with human supervision, to correct typos and other minor errors in the text.

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